

PROBLEM STATEMENT

Algebraic Simplification & Chain Rule

Find the derivative with respect to x of the function:

$$y = \sqrt{\frac{2x}{x^3 - 2x}}$$

STEP 1 প্রদত্ত ভগ্নাংশীয় বর্গমূল ফাংশনটি উল্লেখ করি (Given Radical Rational Function)

$$y = \sqrt{\frac{2x}{x^3 - 2x}}$$

STEP 2 অন্তরীকরণের পূর্বে সাধারণ চলক x বর্জন করে সরলীকরণ করি (Algebraic Simplification)

$$y = \sqrt{\frac{2x}{x(x^2 - 2)}} = \sqrt{\frac{2}{x^2 - 2}} = \sqrt{2}(x^2 - 2)^{-1/2}$$

STEP 3 উভয়পক্ষে x -এর সাপেক্ষে অন্তরীকরণ করি (Differentiating w.r.t. x)

$$\frac{dy}{dx} = \sqrt{2} \cdot \frac{d}{dx} [(x^2 - 2)^{-1/2}]$$

STEP 4 শক্তি বিধি ও চেইন রুল প্রয়োগ করি ($\frac{d}{dx}[u^n] = nu^{n-1} \frac{du}{dx}$)

$$\frac{dy}{dx} = \sqrt{2} \cdot \left[-\frac{1}{2}(x^2 - 2)^{-3/2} \cdot \frac{d}{dx}(x^2 - 2) \right] = -\frac{\sqrt{2}}{2}(x^2 - 2)^{-3/2} \cdot (2x)$$

STEP 5 লব ও হরের ধ্রুবক ২ কাটাকাটি করে চূড়ান্ত অন্তরজ পাই (Final Cancellation & Form)

$$\frac{dy}{dx} = -\frac{\sqrt{2}x}{(x^2 - 2)^{3/2}} = -\frac{\sqrt{2}x}{(x^2 - 2)\sqrt{x^2 - 2}}$$

FINAL ANSWER $\frac{dy}{dx} = -\frac{\sqrt{2}x}{(x^2 - 2)^{3/2}}$

BOX 1: FORMULATION & SIMPLIFICATION STRATEGY

- **Simplification Principle:** Always factor and cancel common variables inside radicals before applying differentiation rules.
- Direct negative power: $\frac{d}{dx}[c(x^2 - a)^{-1/2}] = -cx(x^2 - a)^{-3/2}$.
- Here, $c = \sqrt{2}, a = 2 \implies \frac{dy}{dx} = -\frac{\sqrt{2}x}{(x^2-2)^{3/2}}$.

BOX 2: COMMON STUDENT TRAPS & PITFALLS

- **Unsimplified Quotient Rule Nightmare:** Applying Quotient Rule directly to $\frac{2x}{x^3-2x}$ results in bulky expressions with higher risk of algebraic errors.
- **Negative Exponent Reduction:** $-\frac{1}{2} - 1 = -\frac{3}{2}$. A frequent error is writing $-\frac{1}{2} - 1 = \frac{1}{2}$.

BOX 3: REAL DOMAIN & DIFFERENTIABILITY ANALYSIS

- **Real Domain:** The expression under the radical requires $\frac{2}{x^2-2} > 0 \iff x^2 > 2 \iff |x| > \sqrt{2}$.

$$\text{Domain} = (-\infty, -\sqrt{2}) \cup (\sqrt{2}, \infty)$$

- Note that $x = 0$ is completely outside the domain of definition.
- **Differentiability:** Infinitely smooth on its domain ($|x| > \sqrt{2}$).

BOX 4: LOGARITHMIC DIFFERENTIATION ALTERNATIVE

- Taking natural logarithm:

$$\ln y = \frac{1}{2} \ln 2 - \frac{1}{2} \ln(x^2 - 2)$$

- Differentiating:

$$\frac{1}{y} \frac{dy}{dx} = 0 - \frac{1}{2} \cdot \frac{2x}{x^2 - 2} = -\frac{x}{x^2 - 2}$$

- Multiplying by $y = \frac{\sqrt{2}}{\sqrt{x^2-2}}$:

$$\frac{dy}{dx} = -\frac{x}{x^2 - 2} \cdot \frac{\sqrt{2}}{\sqrt{x^2 - 2}} = -\frac{\sqrt{2}x}{(x^2 - 2)^{3/2}}$$

BOX 5: AMIE EXAMINATION MARKS DISTRIBUTION (TOTAL: 4 / 5 MARKS)

- **1.0 Mark:** Correct initial algebraic reduction of fraction to $\sqrt{2}(x^2 - 2)^{-1/2}$.
- **1.0 Mark:** Correct power rule derivative $-\frac{1}{2}(x^2 - 2)^{-3/2}$.
- **1.0 Mark:** Accurate inner derivative $\frac{d}{dx}(x^2 - 2) = 2x$.
- **1.0 Mark:** Final clean radical fraction form $-\frac{\sqrt{2}x}{(x^2-2)^{3/2}}$.